



ARCHITECTURE DOCUMENT

Commission Calculation System on AWS

Internal-only, audit-ready platform for commission rules, shipment extraction, calculations, and BI exports

Architecture Recommendation

PATTERN hybrid	PROVIDER aws	SCALE enterprise	SECURITY high
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Key Conclusions

- The recommended architecture follows a hybrid approach.
- Show the end-to-end AWS architecture for an internal commission calculation platform with secure access, data ingestion, calculation, auditability, and BI/finance integrations.
- Planning signals indicate aws, enterprise, high requirements.
- The solution stack covers 7 implementation layer(s) with documented rationale and alternatives.
- The application is private-by-default: users reach the system only through VPN access and an internal Application Load Balancer, while Azure Entra provides SSO and MFA for browser-based authentication.
- CCS keeps business authorization inside the application layer with RBAC and office-level data permissions, so identity is centralized in Entra but access rules remain controlled by the commission platform.
- Amazon RDS for PostgreSQL is the system of record; a read replica is used for Qlik and reporting access to isolate analytics load from transactional processing.

Business Value The architecture separates user access, application services, data persistence, and operational concerns so stakeholders can review capability coverage and delivery readiness.	Technical Value The approved blueprint documents major components, integration paths, technology choices, assumptions, and review outcomes in one traceable package.
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Context and Planning Signals

APPLICATION TYPE

data_analytics

DENSITY

standard

COMPONENTS

20

CONNECTIONS

26

Detected Capabilities

authentication

database

security

governance

Constraints

compliance_sensitive

resilience_required

Functional Requirements

- Azure Entra SSO and MFA for user access
- commission rule maintenance with approval/audit
- shipment extraction from CBX landing-stage data
- commission calculation and manual adjustment workflow
- DN/CN generation and billing cycle closing
- reporting data view for Qlik and finance exports
- data migration from allegro commission rules and shipment history

Non-Functional Requirements

- internal network / VPN only
- least privilege access control
- full audit trail for transactions and admin actions
- encryption in transit and at rest
- multi-AZ resilience and daily backups
- 10 years of historical retention
- support about 100,000 vendor invoices annually

Requirement Coverage

TYPE	REQUIREMENT	MAPPED ARCHITECTURE AREA	STATUS
Functional	Azure Entra SSO and MFA for user access	Azure Entra ID, SSL VPN Access, Edge & Access	Addressed in approved blueprint
Functional	commission rule maintenance with approval/audit	CCS API & Rule Service, Commission Calculation Worker	Addressed in approved blueprint
Functional	shipment extraction from CBX landing-stage data	CBX OI Landing Stage DB, Shipment Extraction Worker, Data Platform	Addressed in approved blueprint
Functional	commission calculation and manual adjustment workflow	Workflow Orchestrator, Commission Calculation Worker	Addressed in approved blueprint
Functional	DN/CN generation and billing cycle closing	DN/CN & Billing Close Service	Addressed in approved blueprint
Functional	reporting data view for Qlik and finance exports	Qlik BI, Navision / Finance System, Data Platform	Addressed in approved blueprint
Functional	data migration from allegro commission rules and shipment history	Legacy allegro System, CCS API & Rule Service, Shipment Extraction Worker, Commission Calculation Worker	Addressed in approved blueprint
Non-functional	internal network / VPN only	Internal ALB	Addressed in approved blueprint
Non-functional	least privilege access control	SSL VPN Access, Edge & Access	Addressed in approved blueprint
Non-functional	full audit trail for transactions and admin actions	External Systems & Users, Edge & Access, CCS Users, Azure Entra ID, CBX OI Landing Stage DB	Addressed in approved blueprint
Non-functional	encryption in transit and at rest	External Systems & Users, Edge & Access, CCS Users, Azure Entra ID, CBX OI Landing Stage DB	Addressed in approved blueprint
Non-functional	multi-AZ resilience and daily backups	External Systems & Users, Edge & Access, CCS Users, Azure Entra ID, CBX OI Landing Stage DB	Addressed in approved blueprint
Non-functional	10 years of historical retention	External Systems & Users, Edge & Access, CCS Users, Azure Entra ID, CBX OI Landing Stage DB	Addressed in approved blueprint

TYPE	REQUIREMENT	MAPPED ARCHITECTURE AREA	STATUS
Non-functional	support about 100,000 vendor invoices annually	External Systems & Users, Edge & Access, CCS Users, Azure Entra ID, CBX OI Landing Stage DB	Addressed in approved blueprint

Architecture Approach

CCS combines an interactive internal portal with batch-oriented shipment extraction, commission calculation, DN/CN generation, and reporting integrations. A hybrid architecture on AWS keeps the user-facing path simple while using workflow orchestration and managed PostgreSQL for auditable, resilient processing and direct SQL access for Qlik and finance.

Primary Flow

FROM	TO	LABEL	PROTOCOL
ccs_users	client_vpn	VPN access	TCP
ccs_users	azure_entra_id	SSO + MFA login	HTTPS
azure_entra_id	internal_alb	OIDC authentication	HTTPS
client_vpn	internal_alb	Private app access	HTTPS
internal_alb	ccs_web_portal	HTTPS	HTTPS
ccs_web_portal	ccs_api_service	UI API calls	HTTPS
ccs_api_service	step_functions	Start workflow	HTTP
step_functions	shipment_extraction_worker	Extract / refresh shipments	HTTP
shipment_extraction_worker	cbx_landing_stage_db	Read shipment + PO data	SQL
shipment_extraction_worker	rds_postgresql_primary	Stage merged data	SQL
step_functions	commission_calculation_worker	Run commission calculation	HTTP
commission_calculation_worker	rds_postgresql_primary	Read / write commission results	SQL

Layout and Presentation Constraints

- left-to-right main flow
- aggregate observability, security, and governance as side channels
- keep diagram to a client-facing architecture view, not code-level detail
- show AWS account or environment boundaries only where they clarify compliance
- limit node count to essential enterprise components

Implementation Rationale

LAYER	CHOICE	RATIONALE	ALTERNATIVES
frontend	Internal web portal on Amazon ECS Fargate behind an internal Application Load Balancer	The requirement is a VPN-only internal application with browser access, so serving the UI from the same AWS private application tier keeps access control, deployment, and audit boundaries simple. It also avoids exposing any public CDN surface.	Amazon S3 + CloudFront for a static SPA, Amazon Amplify for a managed frontend
backend	Amazon ECS Fargate services for CCS API, calculation engine, and workflow workers	The platform needs scheduled processing, manual workflows, and enterprise integrations without managing servers. ECS Fargate fits the internal enterprise workload while keeping deployment and scaling straightforward.	AWS Lambda for some event-driven jobs, Amazon EKS for heavier platform standardization
database	Amazon RDS for PostgreSQL (Multi-AZ)	The requirements explicitly call for direct PostgreSQL access for Qlik, strong relational consistency, and long-lived transaction history. RDS PostgreSQL gives managed backups, encryption, read replicas if needed, and straightforward native SQL access.	Amazon Aurora PostgreSQL, Amazon Redshift for analytics-only copies
queue	Amazon SQS with AWS Step Functions for orchestration	Commission extraction, validation, generation, and closing are workflow-oriented and benefit from decoupled job handling with retries and state tracking. This combination supports resilient batch processing and auditability.	Amazon EventBridge only, AWS Batch for large offline jobs
auth	Azure Entra ID federation via Amazon Cognito or ALB OIDC authentication	The system must use the existing corporate identity provider with SSO and MFA. Federation keeps authentication centralized in Entra while AWS handles application session enforcement.	ALB OIDC directly to Azure Entra, AWS IAM Identity Center with external IdP
infra	Amazon VPC with private subnets, internal ALB, AWS KMS, Secrets Manager, IAM, AWS Backup, and VPC endpoints	The system must be internal-only, encrypted, least-privilege, and resilient. This AWS foundation supports network isolation, secret handling, encryption, and multi-AZ deployment.	AWS App Runner with private access, Amazon EC2 Auto Scaling on private subnets
monitoring	Amazon CloudWatch, AWS CloudTrail, and AWS Config	The requirements call for full audit logging, operational dashboards, and compliance traceability. These services cover application metrics, API/user activity auditing, and configuration drift history.	Amazon OpenSearch for log analytics, AWS X-Ray for deeper tracing

Components and Decisions

Key Design Decisions

- The application is private-by-default: users reach the system only through VPN access and an internal Application Load Balancer, while Azure Entra provides SSO and MFA for browser-based authentication.
- CCS keeps business authorization inside the application layer with RBAC and office-level data permissions, so identity is centralized in Entra but access rules remain controlled by the commission platform.
- Amazon RDS for PostgreSQL is the system of record; a read replica is used for Qlik and reporting access to isolate analytics load from transactional processing.
- Step Functions coordinates the extraction → calculation → DN/CN → close workflow, while ECS Fargate workers handle the heavy processing steps with retries and clear audit boundaries.
- CBX is integrated read-only from the OI Landing Stage mirror database, and allegro migration is modeled as a one-time DMS load into the CCS database without write-back to source systems.
- CloudTrail, CloudWatch, Config, KMS, and Secrets Manager provide the compliance side-channel: encryption, secrets handling, drift visibility, and immutable audit evidence without cluttering the main business flow.

Tiers and Components

External Systems & Users - external

CCS Users (Browser), Azure Entra ID (Microsoft Entra ID), CBX OI Landing Stage DB (CDC mirror of CBX), Qlik BI (Qlik), Navision / Finance System (Navision), Legacy allegro System (allegro)

Edge & Access - infra

SSL VPN Access (AWS Client VPN), Internal ALB (Application Load Balancer)

CCS Application - backend

CCS Web Portal (Amazon ECS Fargate), CCS API & Rule Service (Amazon ECS Fargate), Workflow Orchestrator (AWS Step Functions), Shipment Extraction Worker (Amazon ECS Fargate), Commission Calculation Worker (Amazon ECS Fargate), DN/CN & Billing Close Service (Amazon ECS Fargate)

Data Platform - data

RDS PostgreSQL (Multi-AZ) (Amazon RDS for PostgreSQL), RDS PostgreSQL Read Replica (Amazon RDS for PostgreSQL), AWS DMS Migration Task (AWS Database Migration Service)

Security, Governance & Observability - security

AWS KMS (AWS KMS), AWS Secrets Manager (AWS Secrets Manager), Audit, Logging & Monitoring (CloudTrail / Config / CloudWatch)

Workflow Evidence

analyze_architecture_requirements COMPLETED

2026-06-12T04:02:41Z

Detected data_analytics workload, enterprise scale, high security, provider=aws.

propose_diagram_brief COMPLETED

2026-06-12T04:02:45Z

Recorded diagram brief with 7 functional and 7 non-functional requirements.

propose_tech_stack_gate APPROVE

2026-06-12T04:03:07Z

User approved propose_tech_stack.

propose_tech_stack COMPLETED

2026-06-12T04:03:07Z

Approved technology stack covering 7 layer(s).

propose_blueprint_gate APPROVE

2026-06-12T04:04:49Z

User approved propose_blueprint.

propose_blueprint COMPLETED

2026-06-12T04:04:49Z

Approved hybrid blueprint with 20 nodes, 5 clusters, and 26 edges.

render_diagram COMPLETED

2026-06-12T04:10:00Z

Rendered out.png successfully on attempt 1.

render_diagram COMPLETED

2026-06-12T04:10:59Z

Rendered out.png successfully on attempt 2.

render_diagram COMPLETED

2026-06-12T04:11:52Z

Rendered out.png successfully on attempt 3.

render_diagram COMPLETED

2026-06-12T04:12:14Z

Rendered out.png successfully on attempt 4.

render_diagram COMPLETED

2026-06-12T04:12:40Z

Rendered out.png successfully on attempt 5.

submit_critique PASSED

2026-06-12T04:13:21Z

VERDICT: PASS (no findings — diagram is clean, proceed to finalize)

generate_pdf_report_gate APPROVE

2026-06-12T04:26:20Z

User approved generate_pdf_report.

Tool Summary

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  "tool_counts": {
    "render_diagram": 5,
    "search_icons": 3,
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"submit_critique": 1
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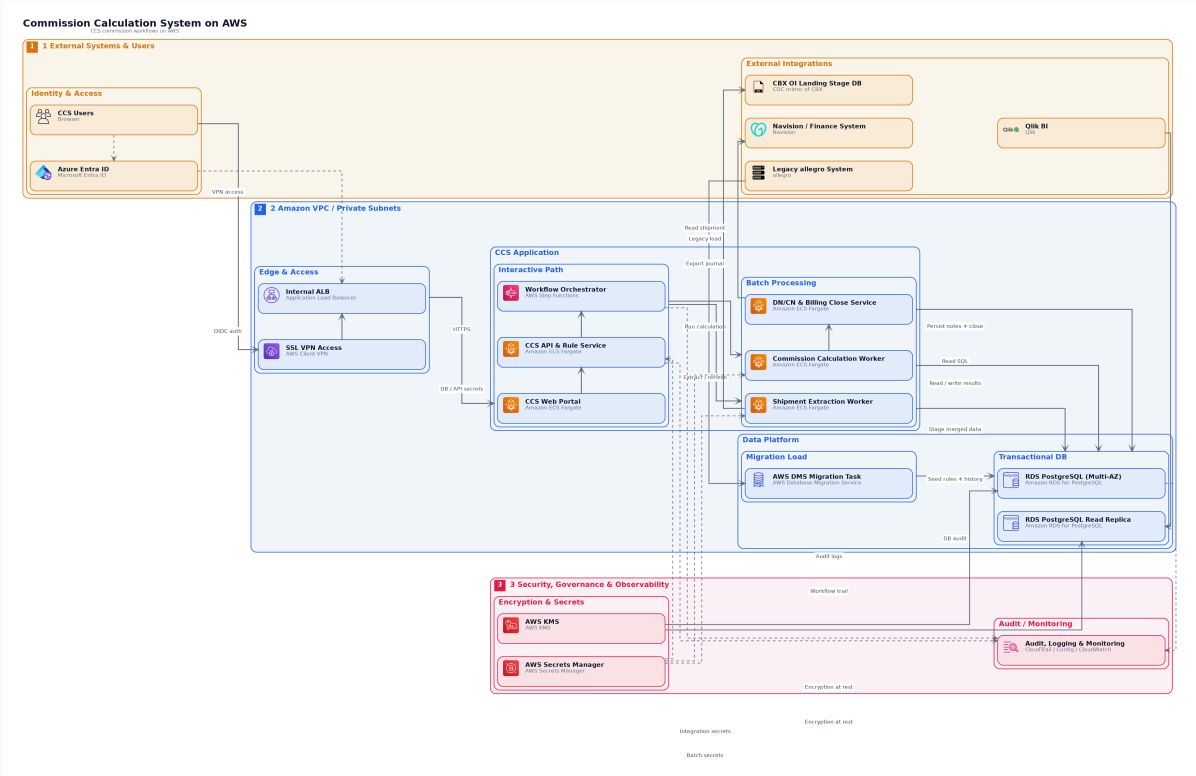
```
}
```

```
}
```

Assumptions, Concerns, and Next Steps

TYPE	DETAIL	RECOMMENDATION
Architecture Concern	Large-scale signal detected; include observability as an aggregated side-channel.	Validate during detailed design.
Assumption	CCS is implemented as a web application with an API/backend tier on AWS	Confirm with stakeholders before implementation.
Assumption	Amazon RDS PostgreSQL is the primary operational datastore	Confirm with stakeholders before implementation.
Assumption	batch/scheduled workflows are orchestrated in AWS rather than manually run	Confirm with stakeholders before implementation.
Assumption	external systems remain outside AWS and are shown as integrations only	Confirm with stakeholders before implementation.
Assumption	multi-account AWS governance is used at least for shared services and production	Confirm with stakeholders before implementation.

END-TO-END CCS AWS ARCHITECTURE



This diagram reflects the approved blueprint and quality review state captured in the preceding sections.

Artifact Inventory

ARTIFACT	DESCRIPTION	SIZE
out.png	Rendered diagram PNG	487725 bytes
out.body.png	Diagram body PNG	534312 bytes
out.drawio	Editable draw.io source	377027 bytes
diagram.py	Generated diagram code	10194 bytes
out.pdf	Client report PDF	1086054 bytes